

Rover wiring: Pi 3B+, ArduSimple RTK2B, XBee-PRO 900HP S3B

GPS module and telemetry radio both connect to the Raspberry Pi 3B+ - GPS via GPIO UART pins, radio via a USB adapter - with an external GPS antenna in and an RF link out to the base station.

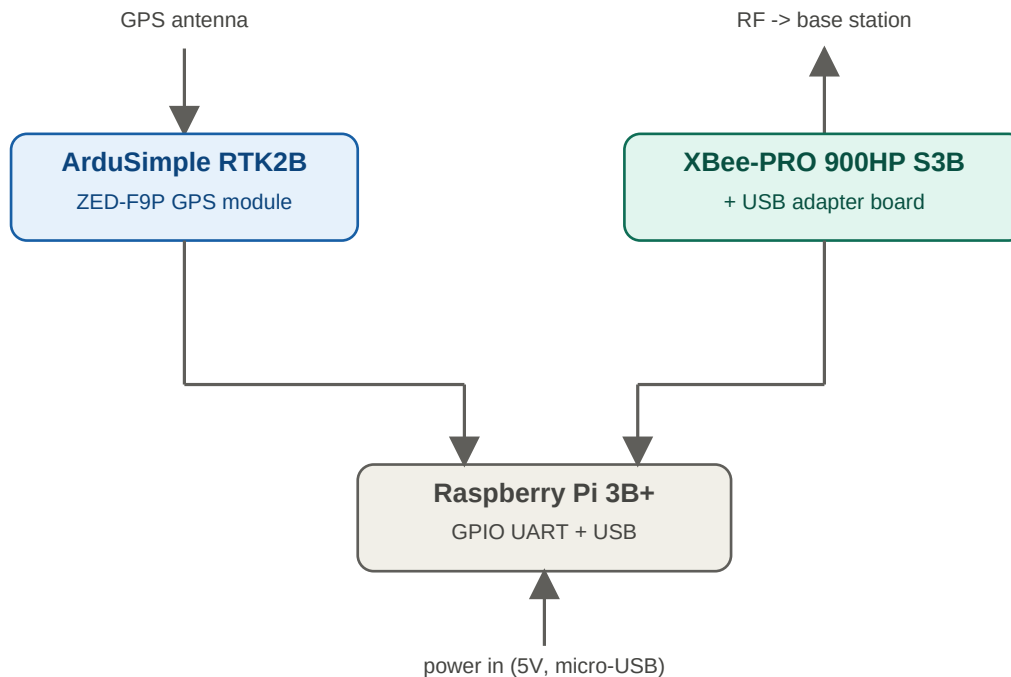


Diagram is a connection map only - see the numbered sections below for exact pins, voltages, and setup commands.

1. GPS: ArduSimple RTK2B ↔ Pi 3B+ (GPIO UART)

ArduSimple RTK2B pin*	Pi 3B+ physical pin	Pi 3B+ signal
TX (GPS transmit)	Pin 10	GPIO15 / RXD
RX (GPS receive)	Pin 8	GPIO14 / TXD
GND	Pin 6 (or any GND pin)	GND
PWR (5V in, if board takes 5V rather than USB)	Pin 2 or 4	5V

*Check your board's silkscreen - header layouts vary across ArduSimple's product revisions.

This is a crossover - GPS's TX goes to the Pi's RX, and vice versa. The ZED-F9P's UART pins are 3.3V logic, matching the Pi's GPIO directly - no level shifter needed.

Pi 3B+-specific gotcha: the 3B+ has onboard Bluetooth, which grabs the Pi's one real hardware UART (ttyAMA0/PL011) by default, leaving GPIO14/15 on the glitchy mini-UART (ttyS0). Fix, before wiring anything electrically matters:

```
sudo raspi-config    # Interface Options -> Serial Port
                    # "login shell over serial" = No
                    # "serial port hardware enabled" = Yes
# /boot/firmware/config.txt (Bookworm+) or /boot/config.txt (older)
```

```
dtoverlay=disable-bt
```

Then disable the Bluetooth services and reboot:

```
sudo systemctl disable --now hciuart bluetooth  
sudo reboot
```

GNSS antenna: connect an active GNSS antenna to the ArduSimple board's own antenna connector (SMA or U.FL, depending on variant) - the board itself supplies the antenna's bias-tee power, no separate wiring.

2. Radio: XBee-PRO 900HP S3B ↔ Pi 3B+ (USB)

The bare XBee-PRO 900HP module uses a 2mm-pitch 20-pin footprint - it can't plug directly into Pi GPIO or a standard header. It needs a carrier/adaptor board (a Digi XBIB-U-Dev, SparkFun XBee Explorer USB, or equivalent) that the module seats into and that exposes a standard USB connector.

- Seat the XBee-PRO 900HP module into the adapter board.
- Connect the adapter's USB cable into any of the Pi 3B+'s 4 USB-A ports (unlike the Pi Zero 2 W this fleet otherwise uses, the 3B+ has full-size USB ports to spare - no hub needed).
- Attach a 900MHz antenna to the XBee's RF connector **before powering it up** - transmitting into an unterminated connector risks damaging the power amplifier.

3. Power

- Pi 3B+: solid 5V/2.5A+ supply via its micro-USB power input.
- The XBee-PRO 900HP can draw close to 1A on transmit in high-power mode - if it's drawing power through the Pi's own USB bus rather than a separately-powered adapter, make sure your supply/cabling has the headroom; a marginal supply will brown out mid-transmit under load.

4. Software side (already documented in the project README)

Once wired, verify GPS baud and enable UBX-NAV-PVT per the existing "GPS configuration" section, then:

```
GPS_PORT=/dev/ttyAMA0 GPS_BAUD=115200 RADIO_PORT=/dev/ttyUSB0 npm run boat
```

(swap /dev/ttyUSB0 for whatever device node the XBee's USB adapter actually enumerates as - check with `ls /dev/ttyUSB*` or `dmesg` after plugging it in).